

MAHMOUD HAMDI EL-SAYED

Data Analyst

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PROFESSIONAL SUMMARY

Data Analyst with hands-on experience designing **end-to-end** analytics solutions — from **relational database architecture** and advanced SQL querying to Power BI dashboards that surface actionable business intelligence across operations, real estate, and marketing analytics domains. Delivered projects processing **100,000+ rows** across multi-source datasets, engineered **Star and Galaxy Schema models**, built Python ETL pipelines, and authored complex **DAX** measures to track revenue, conversion, engagement, and operational performance. Proficient in **SQL Server**, **Power BI**, **Excel**, and **Python** (Pandas, NumPy, Matplotlib), with foundational exposure to dbt and Google BigQuery. Driven to transform raw data into clear decisions.

TECHNICAL SKILLS

SQL Server (T-SQL) • Power BI • Excel (Advanced) • Python • DAX • Power Query (M) • Star / Galaxy Schema • ETL • Git & GitHub

Data Analysis	Statistical Analysis · Descriptive Analytics · KPI Design · Business Problem Framing
Databases	SQL Server 2022 · T-SQL · Relational Schema Design · 3NF Normalization · Views · Stored Procedures
BI & Modeling	Power BI Desktop · Power Query · DAX · Star Schema · Galaxy Schema · Dim_Date Engineering
Spreadsheets	Excel (Pivot Tables · Power Pivot · Conditional Formatting · Dashboards)
Python	Pandas · NumPy · Matplotlib · Data Cleaning
Version Control	Git · GitHub
Exposure	dbt · Google BigQuery

KEY PROJECTS

Commercial-Mall — End-to-End BI System SQL Server · Power BI · DAX · Galaxy Schem [GitHub]	2026
A full-cycle BI project simulating a real commercial mall — from relational database design and SQL analysis through to a production-grade Power BI executive dashboard.	
- Architected a 15-table normalized relational schema in SQL Server 2022 from scratch, enforcing PK, FK, UNIQUE, CHECK, and DEFAULT constraints to guarantee 100% data integrity across tenancy, financial, and HR domains. - Designed a Galaxy Schema (multi-fact star) in Power BI to preserve grain integrity across 7 separate fact tables — Invoices, Collections, Expenses, Violations, and Evaluations — enabling cross-domain analytics without aggregation errors. - Engineered a semantic SQL analytics layer of 7 reusable views built around specific business questions, including financial health reconciliation, overdue invoice seasonality, and passage-level revenue leakage analysis. - Identified a structural 25% collection gap persisting across 4 consecutive fiscal years (2022–2025), with the Main Entrance zone alone carrying a EGP 2.8M collection shortfall despite 95% peak footfall. - Delivered a 2-page executive Power BI dashboard: Page 1 (Financial & Operations — collection efficiency gauge at 74.73%, zone performance, tenant arrear ranking) and Page 2 (HR — attendance compliance at 83.53%, sanction/bonus breakdown, individual productivity ranking). - Flagged critical tenant risk cases including a closed tenant (Trend House) carrying unresolved EGP 239,850 in liabilities — a direct consequence of absent exit-settlement protocols.	

A marketing intelligence initiative built to support data-driven decision-making for Gomla Market's social media team—analyzing owned performance across Facebook, Instagram, and TikTok, while using public Carrefour data as an external benchmark to better understand audience behavior and validate findings.

- Built a **Python ETL pipeline** using Apify to collect and preprocess **4,768+ social media posts** across Facebook, Instagram, and TikTok, including a custom bilingual (Arabic/English) regex-based text-cleaning pipeline for mixed-language content.
- Designed a **Star Schema** data model in Power BI with 3 platform-specific fact tables and 2 conformed dimensions (Dim_Calendar, Dim_Brand), and developed **30+ DAX measures** to support a multi-page marketing intelligence dashboard.
- **Used publicly available Carrefour social media data** as an external industry benchmark to validate audience-behavior patterns and strengthen insights derived from Gomla Market's performance, serving as a learning reference rather than a competitive scorecard.
- **Identified seasonal and event-driven content** (Ramadan, Eid, Back-to-School) as the most consistent driver of engagement, while highlighting posting-time patterns that require further validation before adoption into the content calendar.
- **Produced a six-part executive analysis** including prioritized recommendations, KPIs, implementation timelines, and action owners, providing a practical roadmap to improve content efficiency, platform allocation, and audience engagement.

Comprehensive Power BI analytics platform built across 5 relational tables (9,000+ rows each) covering \$1.54B in revenue, 2,000 units sold, and 5,000 client visits across 2023–2025.

- Built a **Star Schema** data model integrating 2 fact tables (Sales, Visits) and 3 dimension tables, with a fully dynamic DAX-generated Dim_Date spanning the exact dataset range — ensuring clean time-intelligence with no hardcoded boundaries.
- Authored core **DAX KPIs** including Conversion Rate (DIVIDE), Agent Average Performance (AVERAGEX), and Avg Price per SQM — enabling agent-level and property-type benchmarking across 3 fiscal years.
- Diagnosed a **17% YoY drop** in Miami lead/referral rates and identified Q1–Q2 2025 sales stagnation, surfacing extended lead-to-sale cycles as the root cause of revenue underperformance.
- Exposed a wide **agent performance gap**: top 15 agents drove a disproportionate share of total revenue while bottom-tier agents recorded conversion efficiency as low as 17% — with a clear pattern showing high performers specialize in villa and premium-tier closings, not just volume.
- Quantified the 2024 buyer cohort as the highest-**ROI retargeting segment**, providing the business with a precision re-engagement strategy to grow Customer Lifetime Value faster than cold acquisition.
- Delivered a 4-page interactive **Power BI dashboard** covering executive performance review, property portfolio trends, agent leaderboard, and client purchasing behavior — with city-level revenue breakdowns (\$335M New York · \$330M Miami · \$298M Chicago).

CERTIFICATIONS

▪ Microsoft Power BI Data Analyst Professional Certificate	(8 Courses Microsoft)	[View Credential]
▪ IBM Data Analyst Professional Certificate	(11 Courses IBM)	[View Credential]
▪ Google Data Analytics Professional Certificate	(9 Courses Google)	[View Credential]
▪ Data Analysis with SQL & Python	(10+ Courses DataCamp)	[View Credential]

EDUCATION & LANGUAGES

Languages: Arabic — Native | English — Professional Working Proficiency